

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda · Remote Worldwide

PROFESSIONAL SUMMARY

Full-stack engineer with 6+ years of commercial experience across **React.js** and **Node.js**, building web and mobile applications from scratch for international clients in the US, Singapore and East Africa. Delivered pixel-perfect **React UIs from Figma designs** under tight agency timelines, and designed and integrated **REST APIs** serving 2M+ records from third-party providers. Led the end-to-end migration of 20+ production applications to Firebase/GCP with zero downtime, cutting infrastructure spend by \$5,000/month, and architected a microservices backend on Apache Kafka, Docker and Kubernetes. Fully remote since 2020, based in Kampala (UTC+3) with full working overlap across Europe. GitHub: github.com/zac-09

CORE COMPETENCIES

React.js

Node.js

TypeScript

REST API Design & Integration

PostgreSQL / SQL

Pixel-Perfect UI from Figma

Greenfield Application Delivery

Docker & Kubernetes

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote

- Led the end-to-end migration of 20+ applications from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Built real-time two-way sync between MongoDB and Firestore using **Node.js** and Google Pub/Sub for message processing.
- Built a 3D visualisation feature for customers using video overlay and the ffmpeg library, increasing buyer conversion.
- Implemented Express Shipping, generating an additional **\$40 revenue per order**.
- Saved the company **\$5,000/month** by migrating all services off AWS; migrated file storage from Amazon S3 to GCS via a Python script.
- Migrated a WebFlow website from a Parse backend to Firebase.

Stack: Node.js, React, Firebase, GCP Pub/Sub, AWS S3/EC2/EBS, Python, ffmpeg, Webflow, JavaScript

Mind2matter.co Jan 2022 – Jul 2022

Software Engineer · US Remote

- Delivered **React UIs from Figma designs** under tight timelines for a US-based agency with demanding clients, reporting directly to the CTO.
- Built backends for DeFi applications using Web3 and Node.js.

Stack: Node.js, React, Firebase, Tailwind CSS, HTML, CSS, JavaScript

MTailor Mar 2022 – Jul 2022

Contractor · US Remote

- Hired via Upwork to execute a complex backend migration from Parse/MongoDB to Firebase Firestore and GCP.
- Built a migration script to read data from MongoDB and write to Firestore after running complex processing logic.
- Wrote documentation for fellow engineers on working with the new Firebase SDKs (Node.js and Python); trained the Ops team on the Firebase Dashboard and backup procedures.

Stack: Node.js, Firebase, Google Cloud Platform, AWS S3/EC2/EBS, JavaScript

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led optimisation of a PWA delivering real-time stock market prices from a Firebase backend.
- Built ad-hoc jobs to query over **2 million Firestore records**, keeping customer prices current from **Morningstar REST APIs**.
- Extended backend **REST APIs** using Firebase Cloud Functions and Express.js hosted on Heroku.
- Added responsive UI pages to the PWA using HTML, **Tailwind CSS**, and React.

Stack: Node.js, React, Firebase, Express.js, Tailwind CSS, HTML, CSS, JavaScript

Software Engineer · Uganda Remote

- Led a team of 4 developers building legal tech systems for NGOs in Uganda, **built from scratch**.
- Architected a microservices backend for FIDA Uganda's case management app using Apache Kafka, **Docker, and Kubernetes**.
- Developed a cross-platform **mobile app (iOS & Android)** for LASPNET using React Native and Expo, with Firebase push notifications.
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda.
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC.

Stack: Node.js, React Native, Apache Kafka, Docker, Kubernetes, NATS Streaming, gRPC, Firebase, JavaScript

PROJECTS

Parse/MongoDB → Firebase Migration Zero Downtime

Migrated 20+ production applications to Firebase/GCP with no service interruption, running both stacks in parallel behind a real-time two-way Pub/Sub sync layer. Cut infrastructure spend by \$5,000/month.

Node.js · Firebase · Firestore · GCP Pub/Sub · MongoDB · Python

Real-Time Market Data PWA REST Integration

Kept customer-facing stock prices current across 2M+ Firestore records by integrating third-party Morningstar REST APIs, with responsive React + Tailwind pages and Express.js Cloud Function endpoints.

Node.js · React · Tailwind CSS · Express.js · Firebase · REST APIs

FIDA Uganda Case Management Platform Built from Scratch

Greenfield microservices platform for a legal-aid NGO, architected and delivered end to end while leading 4 developers, alongside a cross-platform React Native mobile client and a gRPC-backed USSD service.

Node.js · Apache Kafka · Docker · Kubernetes · gRPC · React Native · Expo

EDUCATION

BSc Computer Engineering — Makerere University

Aug 2018 – Sep 2022

Kampala, Uganda

SKILLS

Frontend: React.js, Tailwind CSS, HTML, CSS, React Native (Expo)

Backend: Node.js, Express.js, REST APIs, gRPC, Apache Kafka, microservices **Data:** SQL, MongoDB, Firestore, Google Pub/Sub

Cloud & DevOps: GCP, Firebase, AWS (S3/EC2/EBS), Docker, Kubernetes, Linux, Git **Languages:** JavaScript, Python, Go, Rust

Spoken: English (fluent), Kiswahili